REPORT

MECHANICAL, DOOR, PNEUMATICS & AC SYSTEMS IN DELHI METRO

Tenure: 8 weeks

Submitted By

Shreya Narayanan

(2021UMV7659)

Mechanical Engineering with specialization in Electric Vehicles, Netaji Subhas University of Technology, Dwarka, Delhi - 110078

Under the Supervision of

Mr. Amit Kumar Jain, Director, Operations and Services

Mr. Vikas Kaushik, SSE, Mechanical

Mr. Amit Pandit, SSE, Pneumatic

Mr. Ashish Kumar Gaur, SE, AC

Mr. Vikram Yogi, SE, Door



DELHI METRO RAIL CORPORATION LIMITED

Metro Bhawan, Fire Brigade Lane, Barakhamba Road, New Delhi - 110001

ACKNOWLEDGEMENT

The internship opportunity I had with Delhi Metro Rail Corporation Limited (DMRC), New Delhi, was a great platform of learning and professional development. Therefore, I consider myself very lucky individual to be a part of it. I am grateful that I had an opportunity to meet wonderful people and professionals who led me through this period and helped me to grow on academic as well as personal level.

It's my radiant sentiment to place on record my best regards and deepest sense of gratitude to **Mr. Amit Kumar Jain, Mr. Vikas Kaushik, Mr, Amit Pandit, Mr. Ashish Kumar Gaur and Mr. Vikram Yogi** for their skilful guidance which has been extremely valuable for my study both theoretically and practically.

Finally, I would like to thank the entire team of DMRC who had been in throughout continuous support and guidance without whom this project wouldn't be a success. I am deeply grateful for pearls of wisdom shared during the tenure of this project.

I perceive this opportunity as a big milestone in my career development. I will strive to use gained skills and knowledge in the best way possible. I will also continue to work on their improvement in order to attain desired career objectives. Hope to continue cooperation with all of you in the future, sincerely.

Shreya Narayanan

(2021UMV7659)

(Mechanical Engineering with specialization in Electric Vehicles)

(Netaji Subhas University of Technology)

OBJECTIVE

The primary objective of the industrial training in DMRC is to get exposure to actual working environment and enhance my knowledge and skills about what I have learned during my engineering course. The main aim of my training in DMRC is to attain knowledge on the Mechanical, Door, Pneumatic and AC System as well as various other systems used in DMRC. The study of Mechanical, Door, Pneumatic and AC System includes various aspects such as the study of Suspension System, Braking System, Traction System, DCU, Isolation Valves etc.

The secondary objective is the brief study of Depot Layout and various other systems and subsystems used in Delhi Metro like Bogies of metro, train formation of metro, HVAC of metro, door system of metro, etc. the aim is also to understand the function of different bodies like PPIO, etc

During my training, I would not only like to get hands-on experience on technical practices but also would like to observe management practices practiced in DMRC and interact with employees and fellow workers.

ABSTRACT

During my training at DMRC, I was explained in detail about the various systems in Delhi Metro, it's components and its functions. My training was on the subject of Mechanical, Door, Pneumatic and AC System.

The bogie, a part of the Mechanical system, is an H shaped structure that supports the load of the car and the passengers, also various systems are mounted on the bogie like suspension system, braking system etc.

Pneumatic systems plays a major role in the working of the train like applying brakes, blowing horn and raising the pantograph while the train is in dead condition.

Doors are the only way for passengers and driver to get out of the train. The doors are electrically controlled and DCU server as the brain for its efficient working.

The AC system is based on the VCR cycle. The temperature inside the train is maintained at room temperature for the comfort of people travelling in it.

Effort is made to explain these systems in an organized way.

LIST OF ABBREVIATIONS

S.No.	Abbreviation	Expanded Form
1.	DMRC	Delhi Metro Rail Corporation
2.	BCU	Brake Control Unit
3.	BECU	Brake Electronic Control Unit
4.	RS	Rolling Stock
5.	IBL	Inspection Bay Line
6.	SBL	Stabling Bay Line
7.	DCU	Door Control Unit
8.	DCC	Depot Control Centre
9.	RSS/TSS	Receiving/Transmitting Sub Station
10.	PPIO	Progression Planning Investigation Officer
11.	OHE	Over Head Equipment
12.	DT/DM	Driving Trailer/Driving Motor
13.	M	Motor
14.	T	Trailer
15.	ATP	Automatic Train Protection
16.	ATO	Automatic Train Operation
17.	BIC	Brake Isolation Cork
18.	MR	Main Reservoir
19.	TIMS	Train Integrated Management System
20.	TCMS	Train Control & Management System
21.	BP	Brake Pipe
22.	WSP	Wheel Slide Protection
23.	VCC	Vacuum Circuit Breaker
24.	HVAC	Heating, Ventilation, and Air Conditioning
25.	PT	Potential Transformer
26.	C/I	Converter/Inverter
27.	SIV	Static Inverter

CONTENTS

Sr. No.	Topic	Page No.
	Introduction to DMRC	8
1.	Introduction to Metro	9
1.1	Benefits of Delhi Metro	9
1.2	Rolling Stock	9
1.3	Administrative Structure	10
2.	Plant Layout	11
2.1	Najafgarh Depot	11
2.2	Najafgarh Depot (Line Diagram)	11

3.	Bogie	12
3.1	Bogie Configuration	13
3.2	Suspension System	13
3.3	Wheelsets	14
3.4	Wheel Mounted Brake Disc	15
3.5	Power Bogie	15
3.6	Comparison of RS-1,RS-2,RS-10 Bogie	16
4.	Pneumatic and Brakes	16
4.1	Pneumatic Air Distribution System	16
4.2	Braking	18
4.3	Braking Mechanism	18
4.4	Types of Braking based on Mechanism	18
4.5	Types of Brakes in Train based on their use	19
4.6	Schematic Diagram of Braking	19
4.7	Block diagram of braking	20
4.8	Flow diagram of braking	20
4.9	Brake Pipe Diagram	21
4.10	WSP	21
4.11	Service Brake unit	21
4.12	Parking Brake Unit	21
4.13	Brake Units	22
4.14	BECU	22
4.15	BCU	23
4.16	Isolation Valves	24
5.	Air Conditioning	24
5.1	Equipment Location	26
5.2	Cab and Car Air Conditioning System	26
5.3	Saloon AC	27
5.4	Compressor	27
5.5	Condenser Fan	28
5.6	Evaporator Coil	28
5.7	Condenser Coil	29
5.8	Filter Drier	29
5.9	Thermal Expansion (Tx) valve	30
6.	Doors	30
6.1	Passenger Doors	30

6.2	General Door Arrangement	31
6.3	Mechanism Overview	32
6.5	Drive Assembly	32
6.6	Automatic Door Locking System	32
6.7	Door Lock Switch	33
6.8	Emergency Egress Device	33
6.9	EDCU	33
7.0	Conclusion	33

LIST OF FIGURES

Fig 1.1	Administrative Structure of Metro	10
Fig 2.1	Najafgarh Metro Depot Plant Layout	11
Fig 2.2	Najafgarh Metro Depot (Line Diagram)	11
Fig 3	Mechanical System	13
Fig 3.1	Bogie Configuration	13
Fig 3.2	Suspension System	13
Fig 3.2	Suspension System	14
Fig 3.2	Secondary Suspension	14
Fig 3.3	Wheelsets	15
Fig 3.4	Wheel mounted brake disc	15
Fig 3.5	Power Bogie	15
Fig 4.1	Main Compressor	17
Fig 4.1	Air Dryer and Filtration Unit	17
Fig 4.1	Main Reservoir	17
Fig 4.1	Auto Drain Valve	18
Fig 4.2	Braking	18
Fig 4.6	Schematic diagram for simple braking	19
Fig 4.7	Block diagram for braking	20
Fig 4.8	Flow diagram for pneumatic braking	20
Fig 4.9	Brake Piping Diagram	21
Fig 4.11	Service Brake Unit	21
Fig 4.12	Parking Brake Unit	22
Fig 4.13	Brake Unit	22
Fig 4.14	Brake Electronic Control Unit (BECU)	23
Fig 4.15	Brake Control Unit (BCU)	23

Fig 5.	Air Conditioning	26
Fig 5.	Air Conditioning	26
Fig 5.1	Equipment Location	27
Fig 5.3	Saloon AC	28
Fig 5.4	Compressor	28
Fig 5.5	Condenser Fan	29
Fig 5.6	Evaporator Coil	29
Fig 5.7	Condenser Coil	30
Fig 5.8	Filter Drier	30
Fig 5.9	Thermal Expansion (Tx) valve	31
Fig 6.1	Passenger Door	30
Fig 6.2	General Door Arrangement	32
Fig 6.3	Mechanism Overview	32
Fig 6.4	Drive Assembly)	32
Fig 6.5	Motor Assembly	32
Fig 6.7	Door Lock Switch	34
Fig 6.8	Emergency egress device	34
Fig 6.9	Electronic Door Control Unit	34

INTRODUCTION TO DMRC

The Delhi Metro Rail Corporation Limited (DMRC) was registered on 3rd May 1995 under the Companies Act, 1956 with equal equity participation of the Government of the National Capital Territory of Delhi (GNCTD) and the Central Government to implement the dream of construction and operation of a worldclass Mass Rapid Transport System (MRTS).

The Delhi Metro has been instrumental in ushering in a new era in the sphere of mass urban transportation in India. The modern Metro system introduced comfortable, air conditioned and eco-friendly services for the first time in India and completely revolutionized the mass transportation scenario not only in Delhi but the entire country.

The DMRC opened its first corridor between Shahdara and Tis Hazari on 25th December, 2002. Subsequently, the first phase of construction worth 65km of Metro lines was finished two years and nine months ahead of schedule in 2005. Since then the DMRC has also completed the construction of another 125km of Metro corridors under the second phase in only four and a half years. At present construction of third phase has almost completed.

Having constructed a massive network of about 373 Km with 271 stations in record time in Delhi, NCR, the DMRC today stands out as a shining example of how a mammoth technically complex infrastructure project can be completed before time and within budgeted cost by a Government agency.

1. INTRODUCTION TO METRO

Metro is like a dream come true for Delhi, a revolutionary change in the city transport. Delhi needs metro system in the first place and it would change things for the betterment of the people who would use it and also for the people living in Delhi by reducing congestion, air pollution, noise pollution and accidents.

The Delhi Metro has also contributed tremendously on the environment front by becoming the first ever railway project in the world to claim carbon credits for regenerative braking. DMRC has also been certified by the United Nations (UN) as the first Metro Rail and Rail based system in the world to get carbon Credits for reducing Green House gas emissions as it has helped to reduce pollution levels in the city by 6.3 lakh tons every year thus helping in reducing global warming.

It has also set up roof top solar power plants at many of its stations. All stations presently under construction corridors are being constructed as green buildings. In the present phase of Delhi Metro's construction, the DMRC is nearing the completion of 160km of Metro lines which has woven a web of Metro corridors along the city's Ring Road besides connecting with many other localities in NOIDA, Ghaziabad, Bahadurgarh and Ballabhgarh.

Apart from providing Delhites with a comfortable public transport option, the Delhi Metro is also contributing significantly towards controlling pollution as well as reducing vehicular congestion on the roads.

1.1. Benefits of Delhi Metro

- Require $1/5^{th}$ energy per passenger as compared to Road-based system.
- Causes no air pollution in the city.
- Causes lesser noise level as compared to Road-based system.
- Reduces travel time (about 50% to 75%).
- It is more reliable, comfortable and safer than Road-based system.
- Occupies no road space if underground and only about 2 m width of the road if elevated.

1.2 Rolling Stock

The **term rolling stock** in DMRC refers to any vehicles that move on a railway. The first wave of Rolling Stock was manufactured by a consortium comprising Hyundai Rotem, Mitsubishi Corporation and Mitsubishi Electric Corporation. Initial sets were built by Rotem in South Korea, with later examples completed in India by public Sector undertaking Bharat Earth Movers Limited (BEML). BEML is also responsible for the manufacturing coaches under technology transfer agreement. The trains have automatic doors, secondary air suspension and brakes controlled by microprocessor.

The Delhi Metro uses rolling stock of two different gauges.

RS-1 and RS-2 uses 1676 mm (5.499 ft) broad gauge rolling stock.

RS-3 and RS-10 uses 1435 mm (4.708 ft) standard rolling stock.

Presently, the Delhi Metro has eleven depots for its existing corridors at Shastri Park, Khyber Pass, Najafgarh, Yamuna Bank, Mundka, Sarita Vihar, Sultanpur, Kalindi Kunj, Ajronda, Mukundpur and Vinod Nagar.

Phase	Gauge	Rolling Stock Version	Extensions	Line
I	Broad Gauge (1676 mm)	RS-1	RS-4 RS-6 RS-13	Red line Yellow line Blue line
II	Broad Gauge	RS-2	RS-5 RS-7 RS-11	Yellow line Blue line
III	Standard Gauge (1435 mm)	RS-3	RS-9	Green line Violet line
III	Standard Gauge	RS-10		Magenta line Pink line Grey Line

Table 1.1. Rolling Stock

Line	Terminals		Track Gauge (mm)
Red Line	Rithala	Shaheed Sthal(New Bus Adda)	1676
Yellow Line	Samaypur Badli	Millennium City Centre Gurugram	1676
Blue Line	Noida Electronic City/ Vaishali	Dwarka Sector 21	1676
			1676
Green Line	Inderlok/	Brig. Hoshiar Singh	1435
	Kirti Nagar		1435
Violet Line	Kashmere Gate	Raja Nahar Singh(Ballabhgarh)	1435
Airport Express (Orange Line)	New Delhi	Yashobhoomi Dwarka Sector 25	1435
Magenta Line	Botanical Garden	Janakpuri West	1435
Pink Line	Shiv Vihar	Majlis Park	1435
Grey Line	Dhansa Bus Stand	Dwarka	1435
Aqua Line	Noida	Greater Noida	1435
Rapid Metro	Sector 55-56	Phase 3	1435

Table1.2. Current Routes

1.3 Administrative Structure

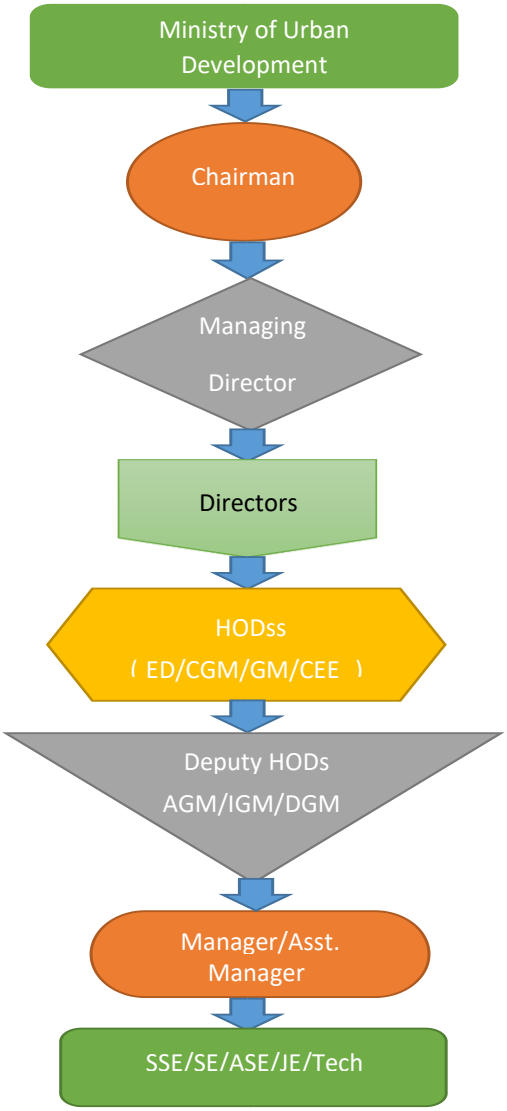
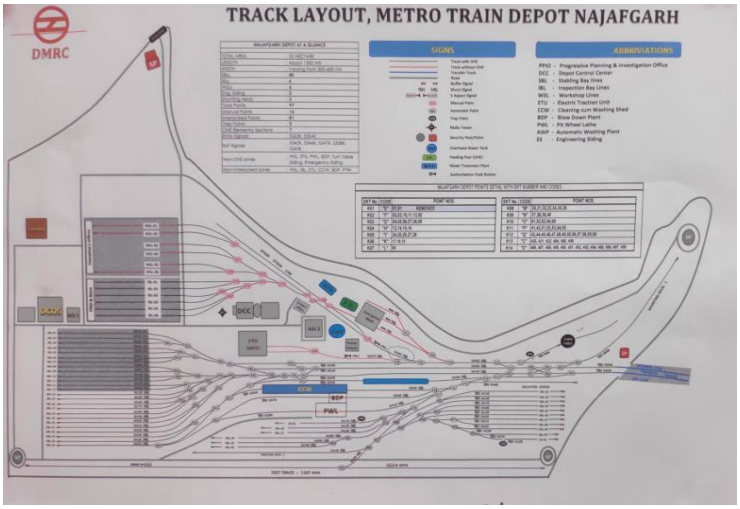


Fig. 1.1 Administrative Structure of Metro

2. Plant layout

2.1 Najafgarh Metro Depot Plant Layout



Najafgarh Metro Depot is situated on one end of Blue line near Dwarka, New Delhi. It serves as a centre for inspection, rescue, testing and maintenance of RS-1 and RS-2 trains running of Blue line between Noida Electronic City/ Vaishali to Dwarka Sector 21. The depot is spread over an area of 52 hectares with 1500m length and varying width of 300m-600m, consisting of 40 SBL, 6 IBL, 6 WSL.

The depot is divided in two sections based on testing and overhauling of different rolling stocks.

One section is for testing and overhauling of RS-1 train and other section is for RS-2 trains. The depot has one metro train testing track of 1167m length.

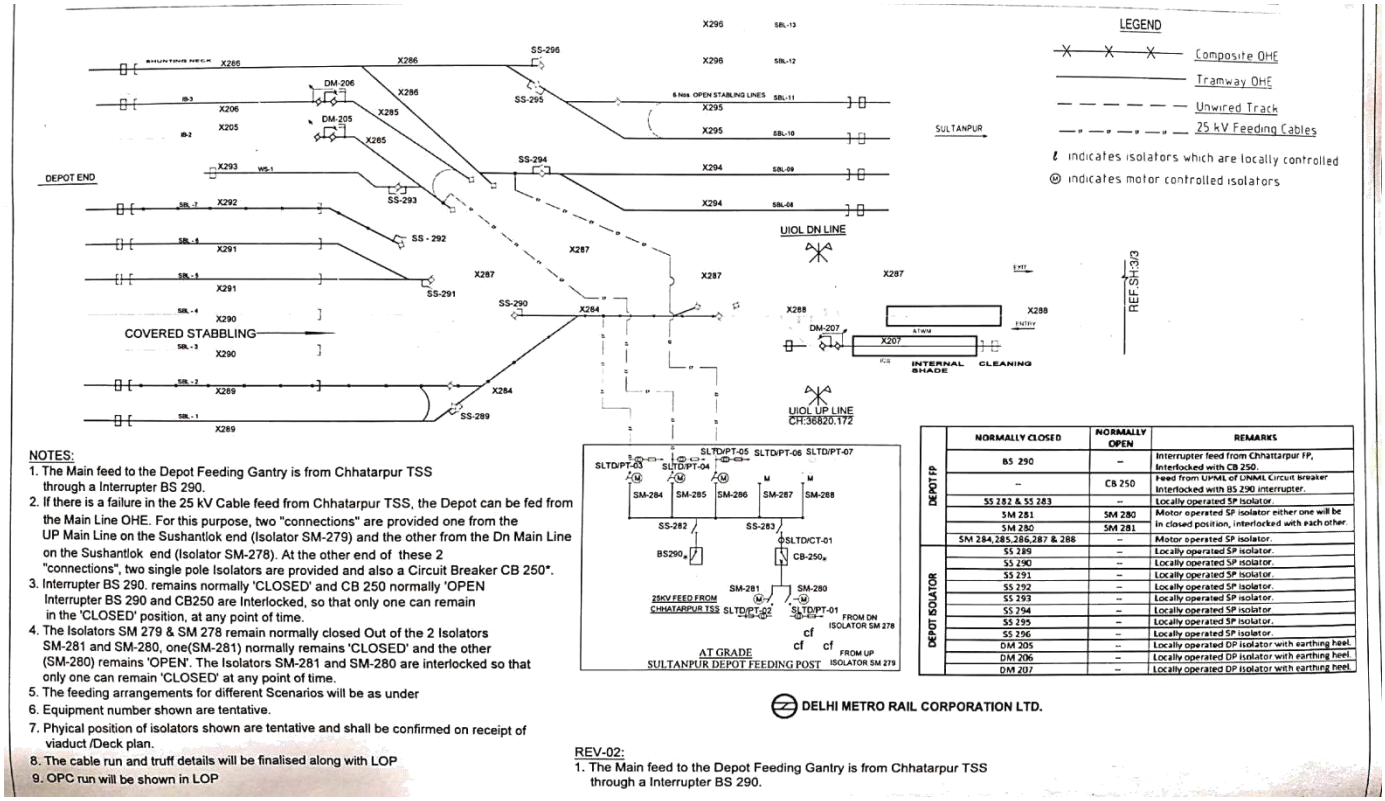
Different signs are used to distinguish different tracks like,

Grey Line: Track with OHE

Red Line: Track without OHE

Blue Line: Transfer Track

2.2 Najafgarh Metro Depot (Line Diagram)

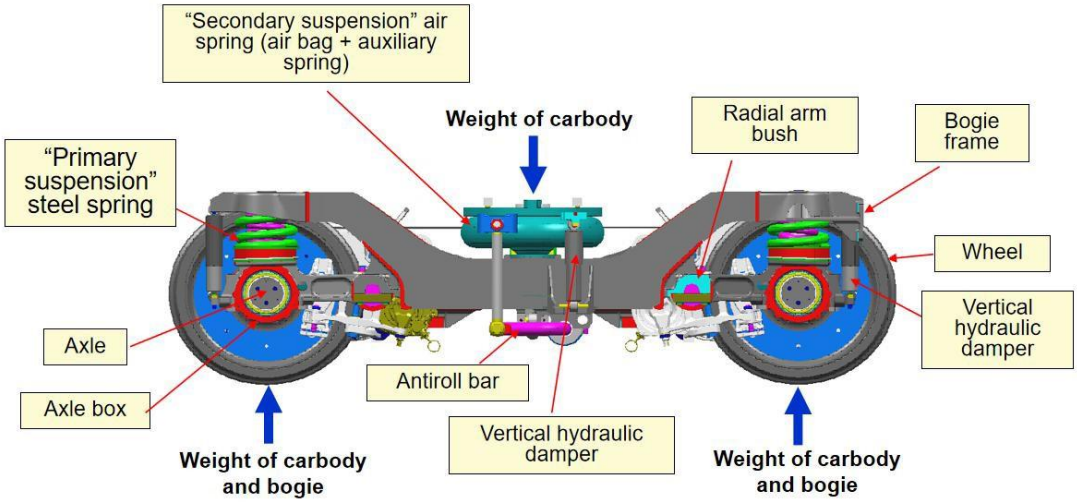


	NORMALLY CLOSED	NORMALLY OPEN	REMARKS
DEPOT FP	BS 290	--	Interrupter feed from Chhatarpur FP, Interlocked with CB 250.
	--	CB 250	Feed from UPML of DNML Circuit Breaker Interlocked with BS 290 interrupter.
	SS 282 & SS 283	--	Locally operated SP isolator.
	SM 281	SM 280	Motor operated SP isolator either one will be in closed position, interlocked with each other.
	SM 280	SM 281	
DEPOT ISOLATOR	SM 284,285,286,287 & 288	--	Motor operated SP isolator.
	SS 289	--	Locally operated SP isolator.
	SS 290	--	Locally operated SP isolator.
	SS 291	--	Locally operated SP isolator.
	SS 292	--	Locally operated SP isolator.
	SS 293	--	Locally operated SP isolator.
	SS 294	--	Locally operated SP isolator.
	SS 295	--	Locally operated SP isolator.
	SS 296	--	Locally operated SP isolator.
	DM 205	--	Locally operated DP isolator with earthing heel.
	DM 206	--	Locally operated DP isolator with earthing heel.
	DM 207	--	Locally operated DP isolator with earthing heel.

3. Mechanical

A bogie is an H shaped structure that supports tractive system, braking system, motor, suspension system etc. Each bogie consists of two axles and four wheels, two bogies comprise one car and combination of one motor car and one trailer car forms a unit. Bogie frame is of two type: 1. Motor bogie frame - The transom area incorporates mountings for the brake equipment, driving gear mounting brackets, traction motor mounting brackets, the mono-link mounting bracket. 2. Trailer bogie frame - The transom area incorporates mountings for the brake equipment, the monolink mounting bracket and ATC beacon antenna mounting bracket is bolted to the transom of front bogie. Both RS-1 and RS-2 bogies are fitted with secondary suspension to ensure good ride quality. Antiroll bar is attached to both the primary suspension to avoid rolling of vehicle while fast cornering and maintaining fixed horizon. The wheels of train are made of mono-block steel using forging process of formation and the axle is fitted with tapered roller bearing. Service brake is fitted in all four wheels of bogie and the parking brake is fitted on two wheels in alternate pattern. Pressurized air is used to control the braking effort on the wheel which is

governed by the BECU(Brake Electronic Control Unit). WSP(Wheel Slide protection) is used to avoid slipping of wheel while heavy braking.



The main function of the bogie is to

1. Support the car body.
2. Allow relative movement between car body and bogie.
3. Ensure stable running.
4. Ensure adequate ride comfort.

Bogie frames are fabricated with steel plates and small insert castings.

Material used - High tensile carbon steel.

Two types of Bogie frames :

1) Trailer bogie frame
2) Power bogie frame

Power bogies are fitted with resiliently mounted traction motors.

There are two stages of suspension:

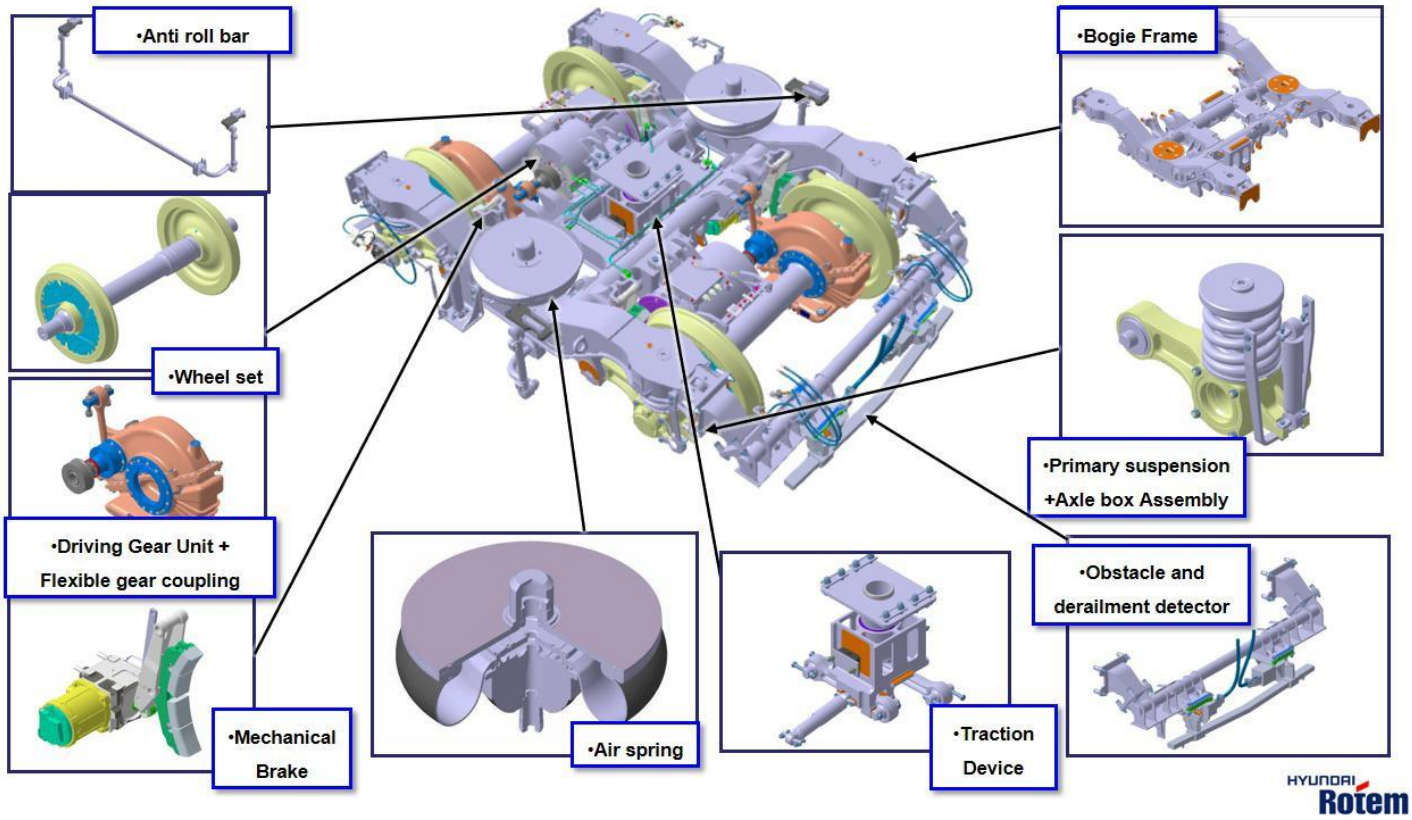
1) Primary suspension
2) Secondary suspension Wheel mounted brake discs.

The wheel sets are made with mono-bloc wheel press fitted onto solid axle.

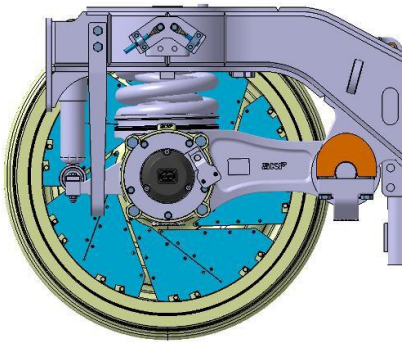
Axle fitted with wheel slide protection (WSP) equipment.

Bogies are fitted with two different types of signaling antennas.

3.1 Bogie Configuration



3.2 Suspension System

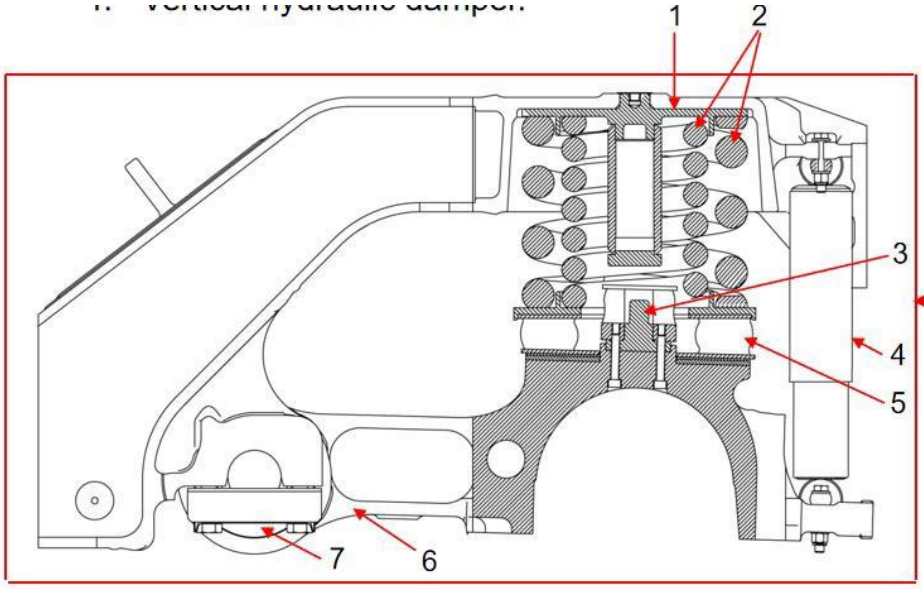


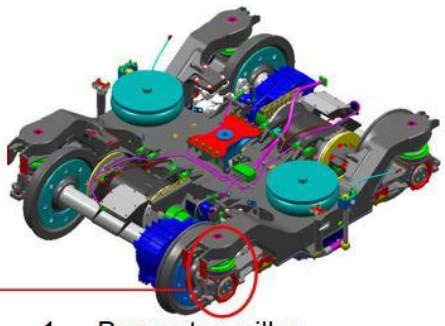
It provides good ride quality to the passenger and reduce vibration and vibration generated stresses.

Primary suspension

The primary suspension has four principal suspension components:

1. Vertical coil springs, mounted in series on a stiff rubber shear pad.
2. Additional progressive vertical rubber spring, arranged in parallel.
3. Resilient radial arm pivot bush.
4. Vertical hydraulic damper.



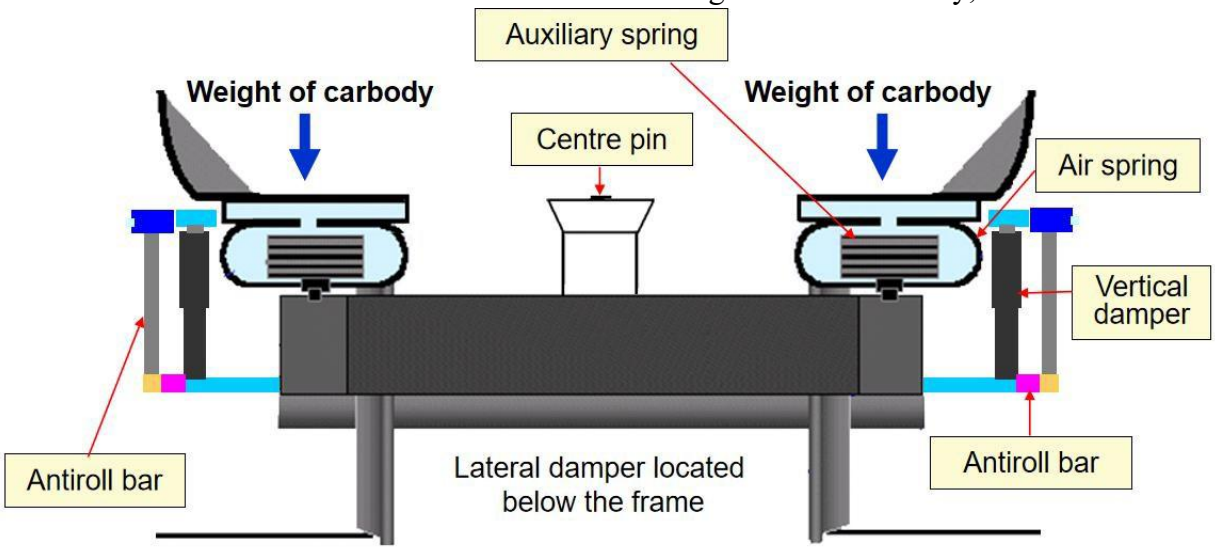


1. Bumpstop pillar
2. Coil springs
3. Hard stop
4. Vertical damper
5. Shear pad
6. Radial arm
7. Radial arm pivot bush

Secondary Suspension

Secondary suspension comprises of :

1. Air spring with auxiliary spring: Air displacement springs support the car body weight and allows the vertical, lateral and rotational displacement between bogie and body. In the event of loss of air pressure, the body is supported by the auxiliary rubber springs, which ensures adequate deflated performance.
2. Antiroll bar : An antiroll bar is connected between the bogie and the car body, to control roll and sway



Schematic diagram of secondary suspension

movements.

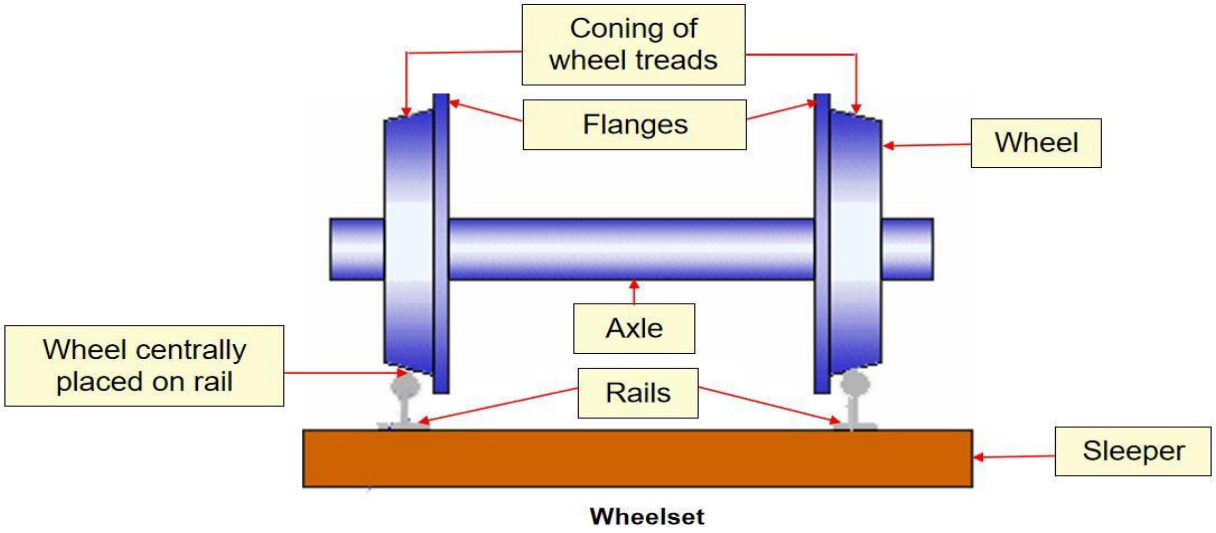
3. Vertical and lateral secondary dampers: Two vertical and one lateral dampers per bogie controls vertical and lateral body oscillations.
4. Centre pin: Braking and traction forces are transferred between the bogie and the car body by a centre pin.

3.3 Wheelsets

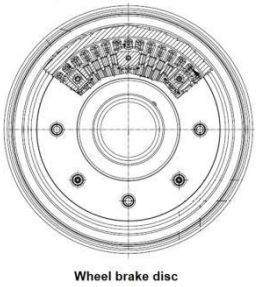
The wheel enables movement of an object across a surface.

An axle is a central shaft for a rotating wheel.

A wheel set is the wheel and axle assembly.



3.4 Wheel mounted brake disc

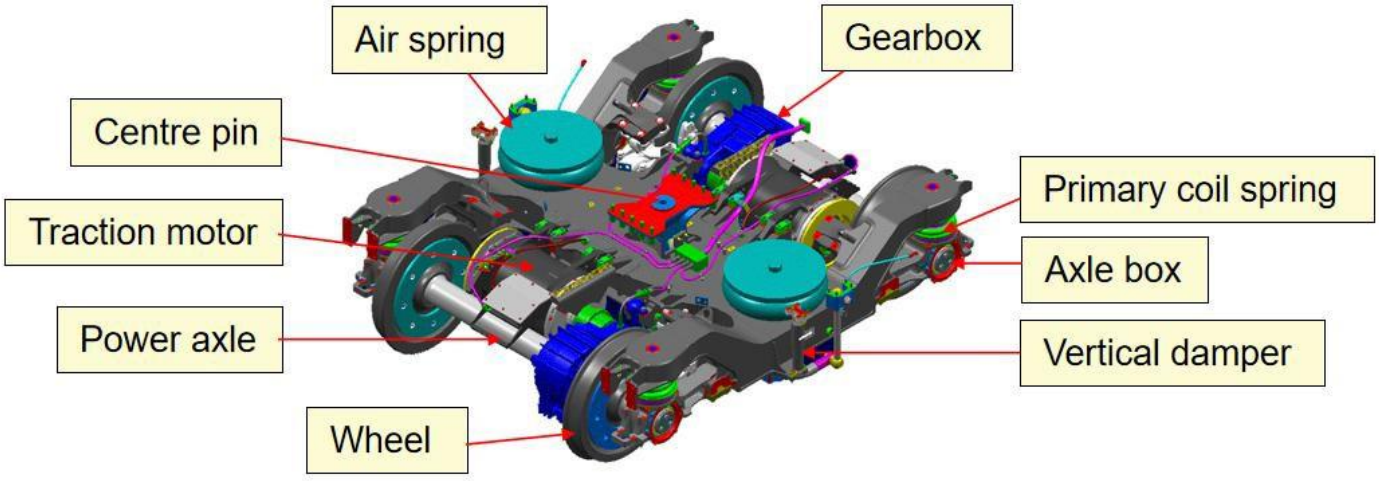


Every wheel is fitted with wheel mounted brake discs (cast iron) bolted onto the wheel web with eight bolts. Disc has fins on the back to help cool the disc.

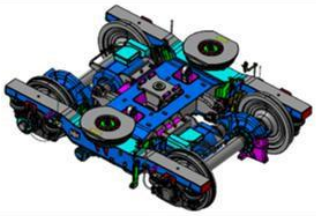
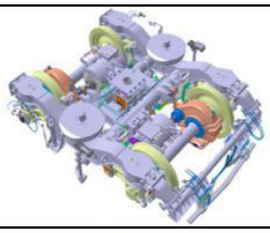
3.5 Power Bogie

Powered bogie consists of:

1. Two power axles, wheel mounted brake discs.
2. Radial arm axle box & primary suspension coil spring, with additional rubber spring.
3. Air spring secondary suspension.
4. Resiliently mounted traction motor.
5. Axle mounted gearbox mounted with resiliently torque.
6. Service brake units and two service/parking brake units with manual release.
7. Axle end earth unit – safety earth.



3.6 Comparison of RS-1, RS-2 and RS-10 Bogie

	RS1	RS3	RS10
			
Track gauge	1,676mm	1,435 mm	1,435 mm
Frame	Conventional H frame (Box transom)	Conventional H frame (Tubular pipe transom)	Conventional H frame (Tubular pipe transom)
Primary suspension	Rubber spring	Rubber spring	Coil spring
Secondary suspension	Air spring + Anti-roll bar	Air spring + Anti-roll bar	Air spring + Anti-roll bar
Traction device	Center Pivot + Mono-link + Lateral damper	Center Pivot + Mono-link + Lateral damper	Center Pivot + z-link + Lateral damper
Brake	Tread Brake	Wheel disc Brake	Tread Brake
Lubrication	NA	Stick type (WFL only)	Spray type (WFL+Check Rail+TOR)
ODD	NA	NA	O

4. Pneumatics

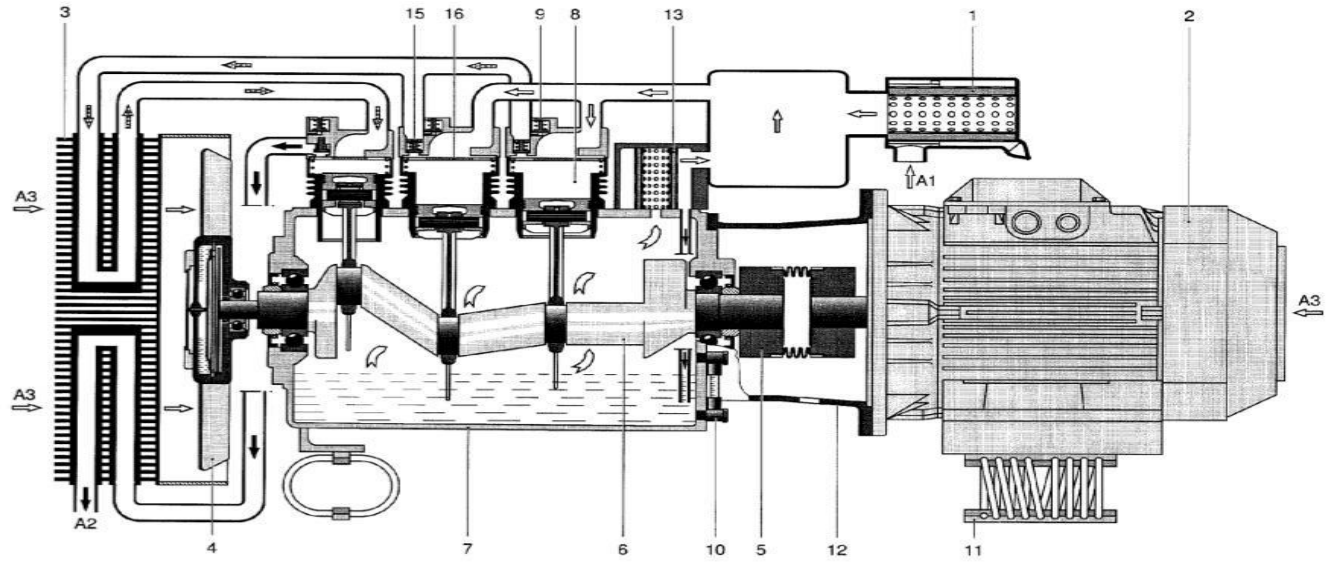
4.1 Pneumatic Air Distribution System

Each unit contains one auxiliary compressor which runs on 110V DC supply and one main compressor (VV120) which is used to charge the reservoirs up to 10bar so that the various mechanisms of the train can run effortlessly. Pressurized air is used:

1. To blow the horn
2. To apply pneumatic brakes
3. To raise pantograph

Main Compressor

The compressor is a cylinder piston compressor powered by a 415 V AC, 50Hz motor and it has a capacity of approximately 900 L/min free air at 10 bar. The main air-compressor unit is controlled by one relay contact of BECU in the software to 8.0 bar for cut in and 10 bar for cut out. The second pressure switch (A9) – set to 7.5 bar for cut in and 10 bar for cut out is provided as hardwired back-up in case of failure of BECU. Based on the electrical signals of the pressure switch A9 and pressure transducer A10 the main motor contactor is controlled. In normal service operation the air compressors of each DT car is controlled individually by the relevant BECU at that trailer car. The BECU in turn communicates with all other BECU installed in the train consist to determine which air compressor should be operated.



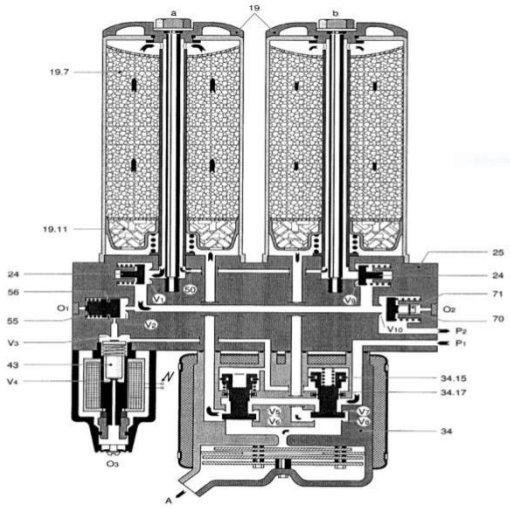
Air intake
Intermediate pressure
High pressure
Oil
Air containing oil

1	Air filter	5	Bellows coupling	9	Safety valve	13	Oil trap
2	Electric motor	6	Crankshaft	10	Oil gauge pipe	15	Delivery valve
3	Cooler	7	Crankcase	11	Spring member	16	Suction valve
4	Fan wheel + viscous coupling	8	Cylinder	12	Intermediate flange		

A1	Air inlet
A2	Air outlet
A3	Cooling air

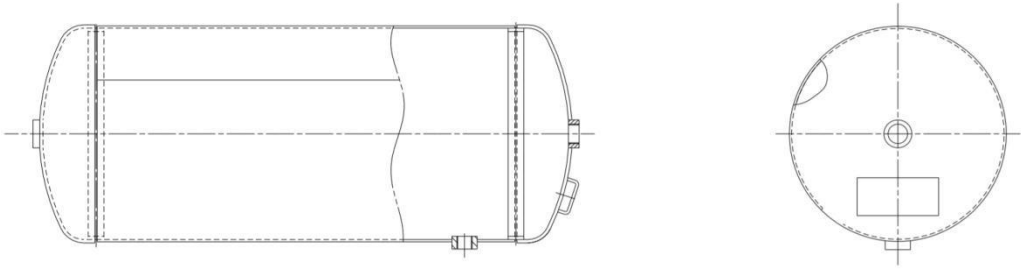
Air Dryer and filtration unit

Moisture in compressed air systems often causes the pneumatic equipment to malfunction and wear prematurely as a result of corrosion and freezing. The compressed air supplied by the main air compressor is filtered by oil separator and dried by the air dryer unit and filtered by the oil micro filter. The air dryer unit consists of two chambers with a adsorptive desiccant. While the main air flow is being dried in one chamber (I), the desiccant of the other chamber is being regenerated by the reversed air flow.



Main Reservoir

Air supply system is provided with main air reservoirs (A7) with a capacity of 200 liters & located in each DT

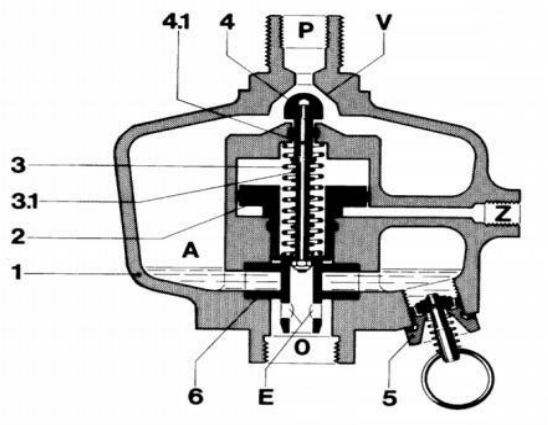


Car

`

Auto Drain Valve

The automatic drain valve is used for automatic drainage (with manual drain also) of condensates from compressed air reservoirs and oil/water separators. The drain valves are operated pneumatically by the magnet valve A17.

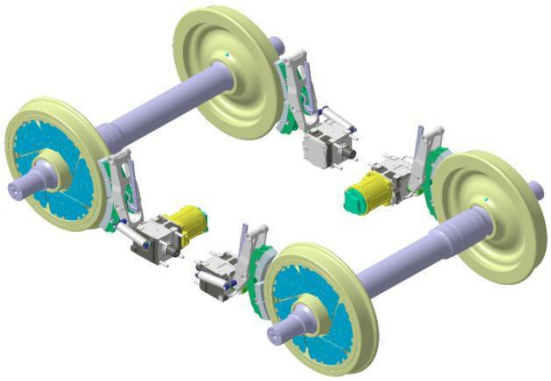


1	Body	**4.1**	Valve stem	**P**	Inlet port
2	Body	**5**	Manual drain valve	**V**	Conical valve seat
3	Spring	**6**	Sealing ring	**Z**	Control air port
3.1	Spring	**A**	Sump	**O**	Exhaust port
4	Valve	**E**	Drain bore		

4.2 Braking

The function of the braking system is to slow or stop the train in motion in adequate stopping distance without wheel slipping and any breakdown.

The brake control system provides an activated train with a safe brake in all situations.



4.3 Working concept of simplest brake system

The pressurized air developed in compressor pushes piston in the brake cylinder. The piston is connected through mechanical linkage to brake shoes rubs on the train wheels. Friction developed by rubbing, slows or stops the train.

4.4 Types of brakes based on their mechanism

1. **Friction brake:** When friction brakes are applied, the kinetic energy of the train will be converted into heat. The friction brake stores the heat in the rotating part (disc brake) during the brake application and then releases it to the air gradually. The disc brake is a device for slowing or stopping the rotation of a wheel.

The disc brake unit is connected to the wheel. To stop the wheel rotation brake pads are forced pneumatically against the disc. Friction causes the disc and wheel to slow or stop the rotation.

The friction brake is a purely elector-pneumatic brake. This means the electronically controlled valves supply a pilot pressure to the relay valve which feeds the brake cylinders with needed pressure.

Service brake, emergency brake and holding brake are achieved by the use of a friction brake.

2. **Spring brake:** It is a mechanical brake. Brake pads are applied to the wheel by the force exerted by spring. The spring activates mechanically.

The parking brake is achieved by the use of spring brake.

3. **Electrical brake:** In electrical breaking the kinetic energy of the train will be converted into electrical energy. The conversion is done by the traction motors, which work as generators during electrical braking. The electrical energy developed can be fed back to the catenary (regenerative braking) or the electrical energy developed can be fed to the resistors (dynamic braking).

Electro-dynamic regenerative brake is achieved by using electrical brake.

4.5 Types of Brakes in Train based on their use

Four types of brakes are used in the metro train:

1. **Service Brake**: Electric Regenerative brake and Electro-Pneumatic (EP) friction brake blending, Load weighed and Jerk Controlled Priority of braking effort;

i. Electric Regenerative brake controlled by C/I of Motor car ii.

EP friction brake controlled by BECU of Driving trailer car

iii. EP friction brake controlled by BECU of Motor car (In the beginning and end of service braking) iv.

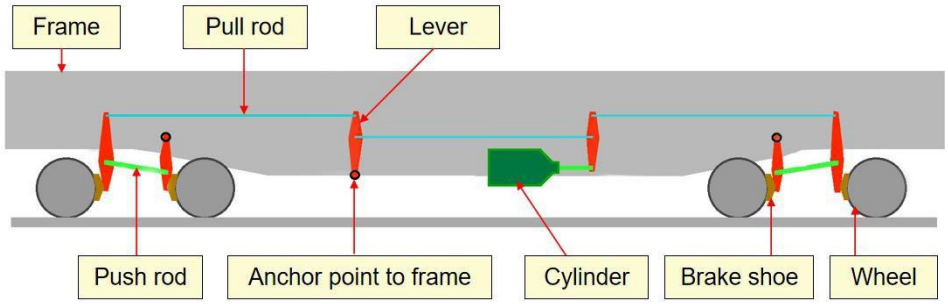
EP friction brake of Driving Trailer car & Motor car (In case of failure of electric brake)

2. **Emergency Brake**: EP friction brake, Load weighed

3. **Parking Brake**: Spring applied, air-release parking brake

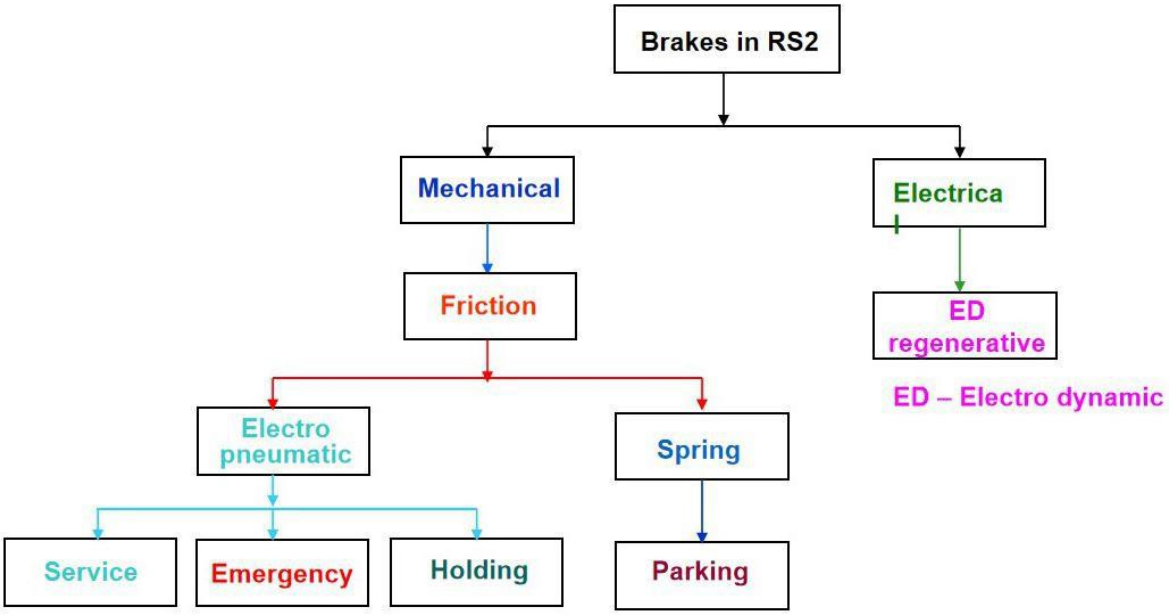
4. **Holding Brake**: EP friction brake Each wheel in the bogie contribute in the overall braking force as all four wheels are provided with brake caliper and then two are used while parking. The actuator of parking brakes are equipped with and extra piston and spring arrangement which is pneumatically loaded.

4.6 Schematic diagram for simple braking

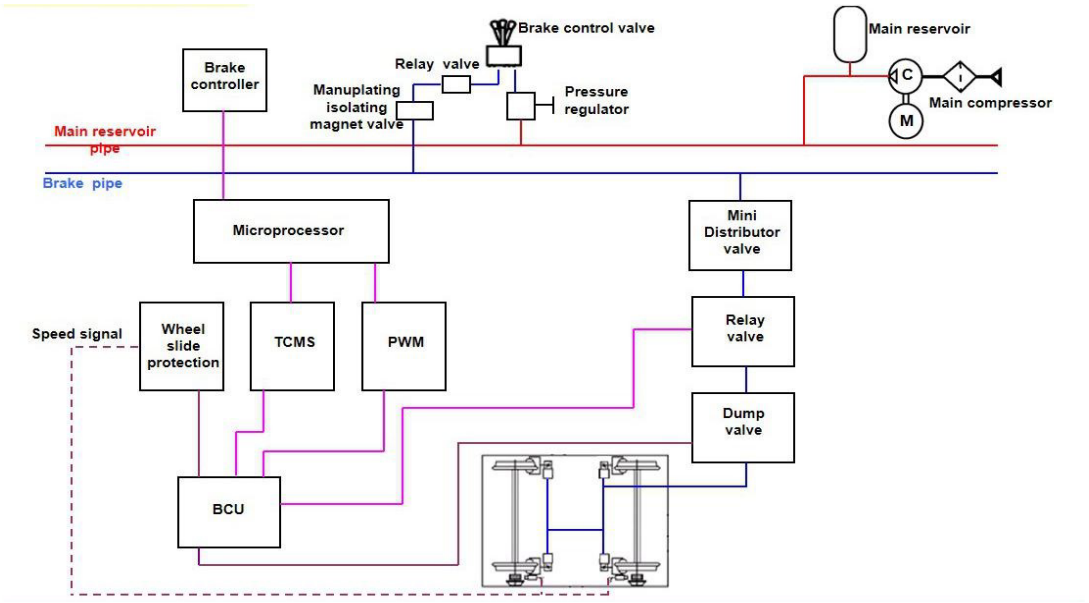


Schematic of simple brake system

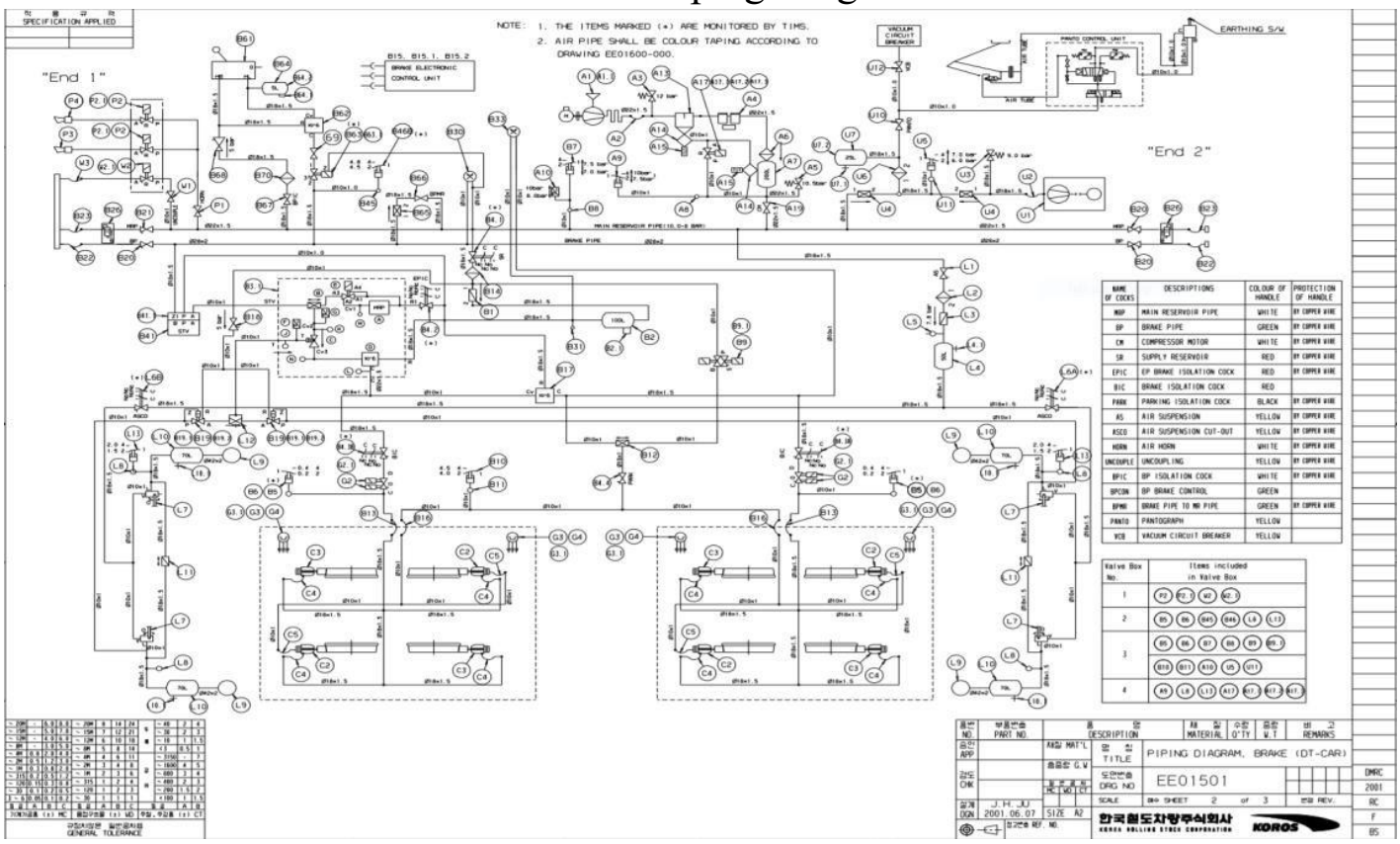
4.7 Block diagram for braking



4.8 Flow diagram for Pneumatic braking



4.9 Brake Piping Diagram

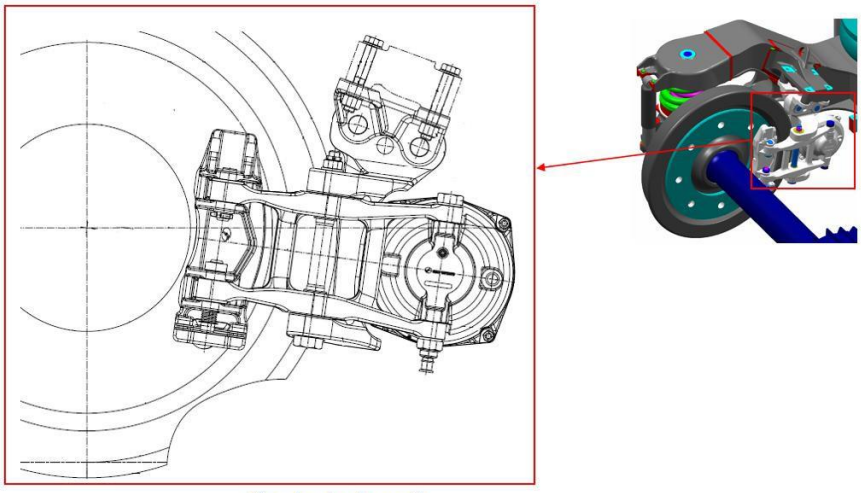


4.10 Wheel Slide Protection (WSP)

The WSP is a system whose purpose is to prevent the wheel sets from locking during a brake application in order to avoid a slide. It minimizes any increase in stopping distance and hence protect protects the wheels and and avoid train sliding.

4.11 Service brake unit

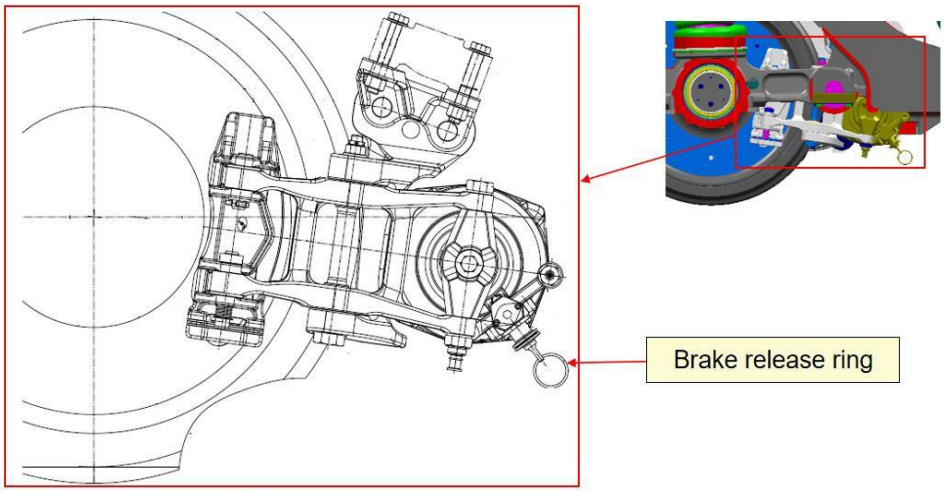
Every bogie is fitted with two service brake units per bogie.



Service brake unit

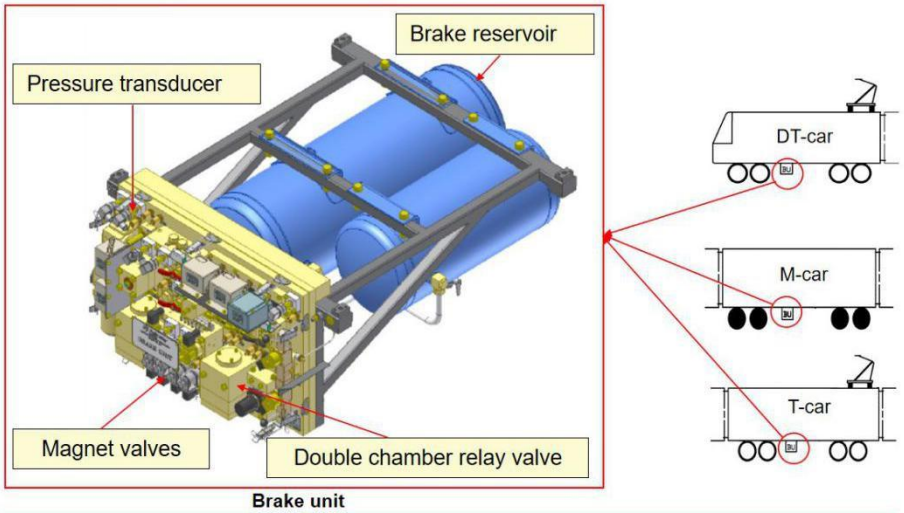
4.12 Parking brake unit

Every bogie is fitted with two combined service/parking brake units per bogie.



Parking brake unit

4.13 Brake unit



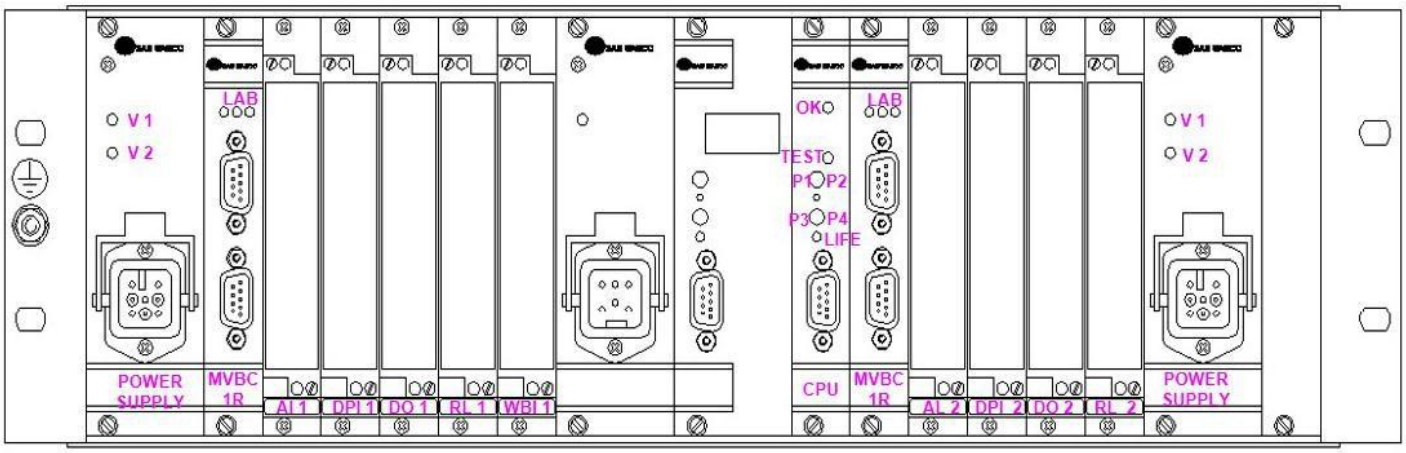
Brake unit

4.14 Brake Electronic Control Unit (BECU)

The electronic rack/brake control unit (BCU) is located in a cubicle above the floor in each car. The electronic rack is used to control the brakes and provide WSP control on that vehicle.

The BECU performs the following functions:

1. Bus connection to TCMS via multi-function vehicle bus (MVB)
2. Weigh braking demand.
3. Jerk control.
4. Automatic brake test.
5. Management of speed signals.
6. WSP functions and testing.
7. Diagnostics.



Electronic rack/BCU

4.15 Brake Control Unit (BCU)

One pneumatic brake control unit is provided in every car. The BCU has following main components a. Analog Converter

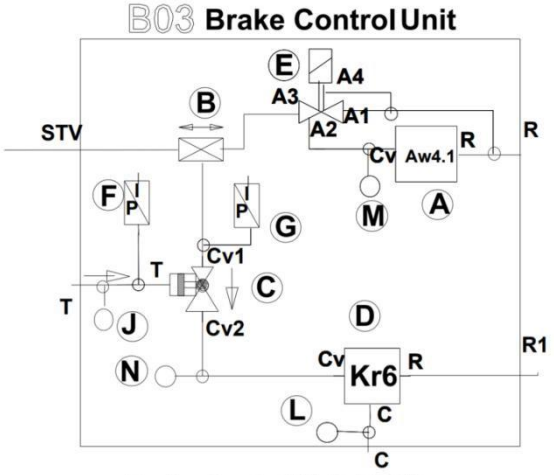
b. Emergency Brake Valve

c. Double Check Valve

d. Pressure Transducer

e. Load Limiting Valve

f. Relay Valve



Brake Control Unit (BCU)

4.16 Isolating valves

Isolation purpose	Description	Physical Position	Quantity in car		
			DT	M	T
Cut out cock	Manual isolation of BP in car end	Coupler area	2	2	2
Cut out cock	Opens the air supply from BP from MRP, during towing with a car with BP	DT-car under frame at front area	1		
Cut out cock (with electric Switch)	Isolates the brake valve from BP	In cab area	1		
	Isolates the brake valve from MRP	In cab area	1		
	Isolates the air supply from MRP to brake unit. All friction brakes and parking brakes in the car are isolated	On the panel in the brake unit	1	1	1
	Isolates EP-brake and emergency brake in car	On the panel in the brake unit	1	1	1
	Isolates the friction brake in one bogie	Inside the passenger saloon area behind lockable cover	2	2	2
Cut out cock	Isolates the friction brake in one bogie	In underframe near each bogie	2	2	2
Cut out cock (with electric Switch)	Isolates the parking brake in the car	On the panel in the brake unit	1	1	1

5. Air Conditioning

The air conditioning in metro systems typically refers to air conditioning systems installed in metro trains. These systems are crucial for maintaining a comfortable temperature for passengers, especially during hot weather.

The AC system is based on the VCR cycle which comprises of the following four components:

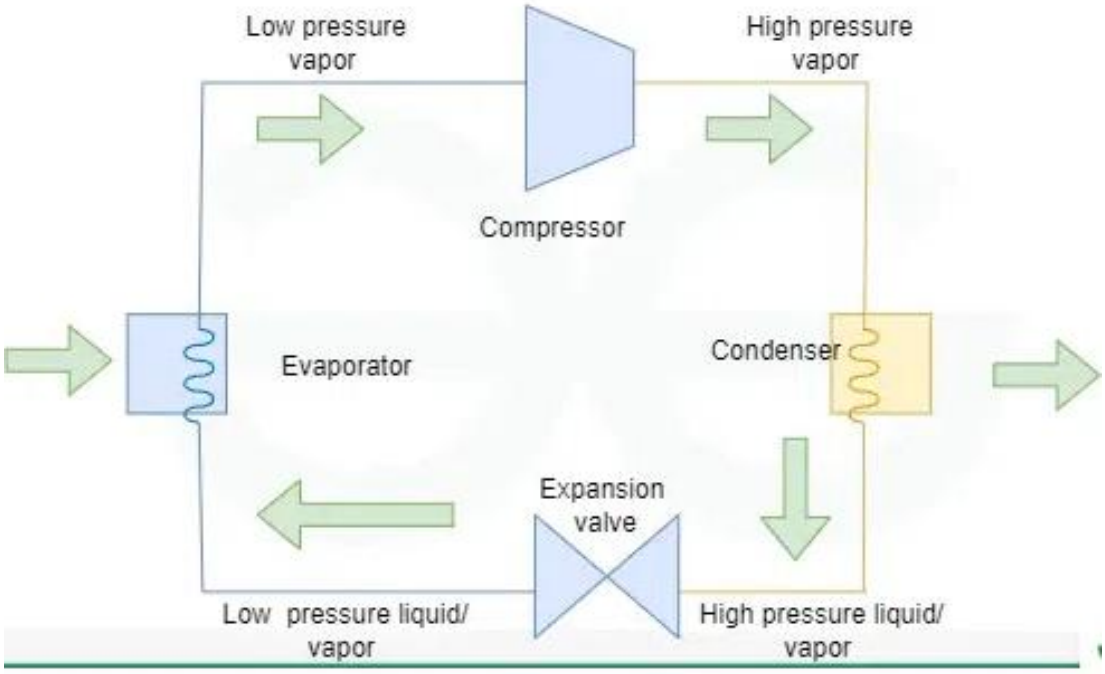
- Compressor

- Condenser

- Expansion valve/throttle valve

- Evaporator

It is a compression process, whose aim is to raise the refrigerant pressure, as it flows from an evaporator. The high-pressure refrigerant flows through a condenser/heat exchanger before attaining the initial low pressure and going back to the evaporator.



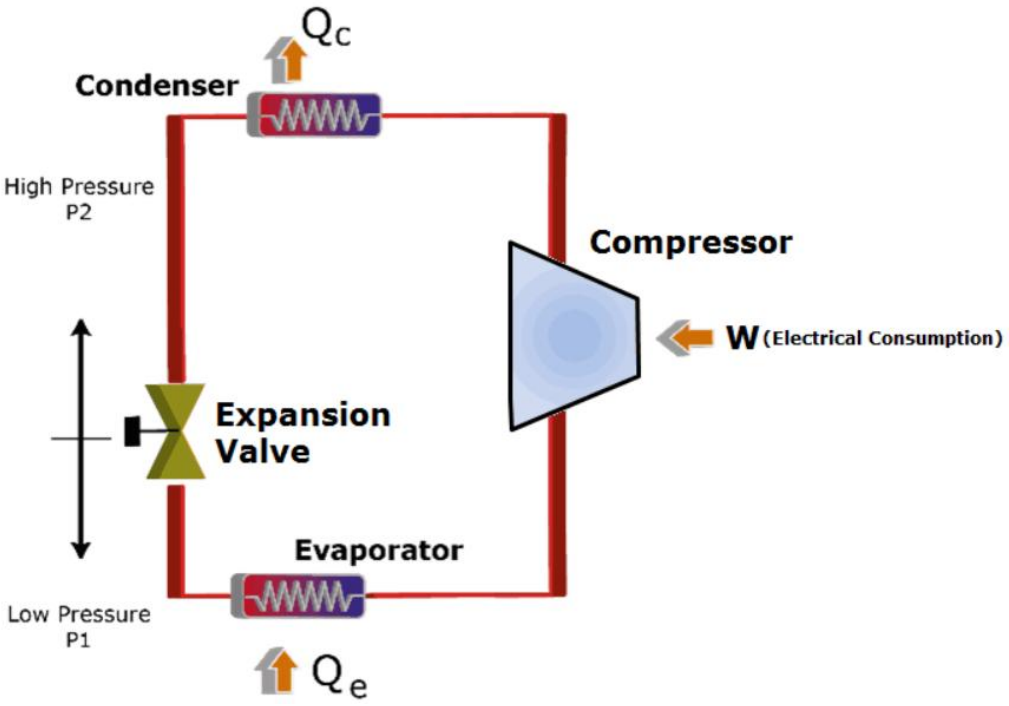
The steps involved are:

1. **Compression:** The refrigerant (for example R-717) enters the compressor at low temperature and low pressure. It is in a gaseous state. Here, compression takes place to raise the temperature and refrigerant pressure.

2. **Condensation:** As the refrigerant flows through the condenser, it is in a constant pressure. The temperature of the refrigerant comes down.

3. **Throttling and Expansion:** When the refrigerant enters the throttling valve, it expands and releases pressure. Consequently, the temperature drops at this stage.

4. **Evaporation:** At this stage of the Vapor Compression Refrigeration Cycle, the refrigerant is at a lower temperature than its surroundings. Therefore, it evaporates and absorbs latent heat of vaporization. Heat extraction from the refrigerant happens at low pressure and temperature.

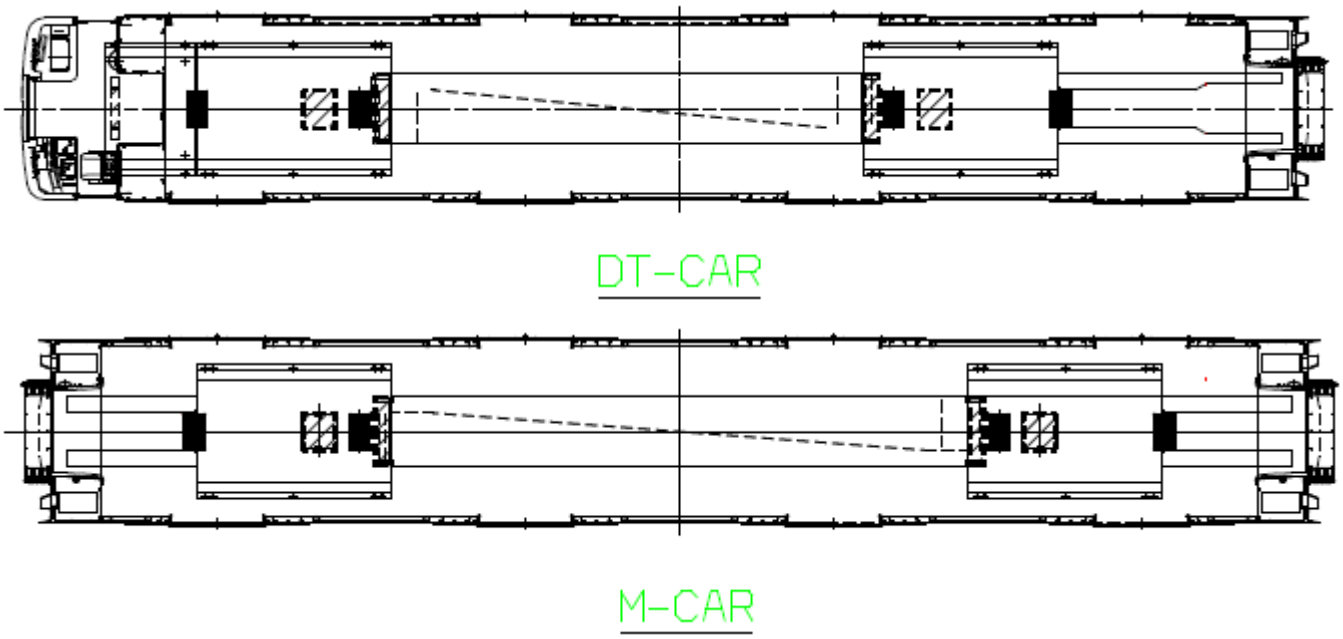


Each saloon car is provided with two air conditioning units providing sufficient cooling capacity even in case of failure of any unit.

Each Saloon Air Conditioning (A/C) unit has a nominal cooling capacity of 41kW under design conditions i.e. 82kW per vehicle.

5.1 Equipment Location

The A/C units are located on both ends of the vehicle roof.

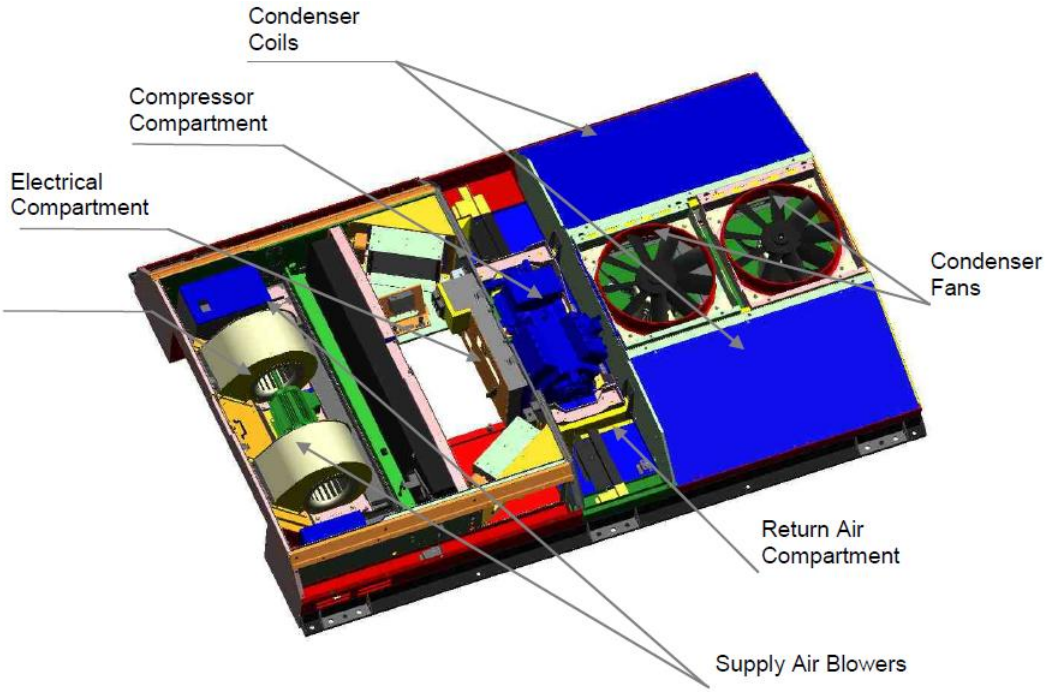


5.2 Cab and Car Air Conditioning System

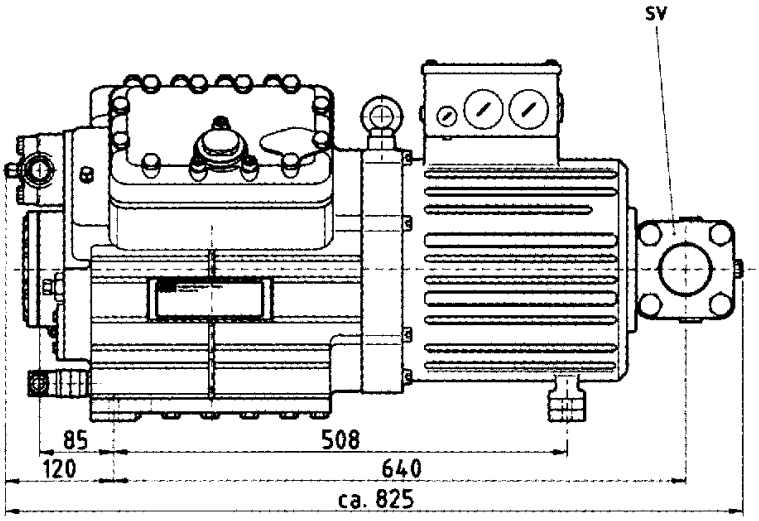
Major Components -

1. Compressor
2. Condenser Coils
3. Condenser Fan
4. Supply Air Blowers
5. Evaporator Coils
6. Return Air Filters
7. Filter Drier
8. Fusible Plug
9. Dual Pressure Control
10. Sight Glass
11. Liquid Line Solenoid Valve
12. Service Access Valves
13. Thermal Expansion (TX) Valve

5.3 Saloon AC



5.4 Compressor



Compressor (Semi Hermetic) is located in the Condenser / Compressor compartment of the A/C unit.

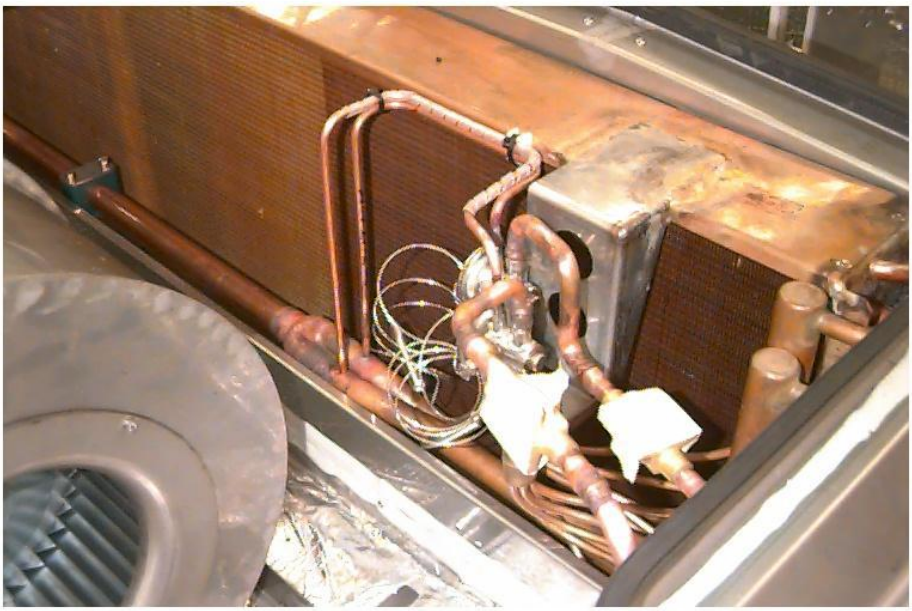
5.5 Condenser fan



Two multi-blade axial fans supply fresh air into unit through condenser coil.

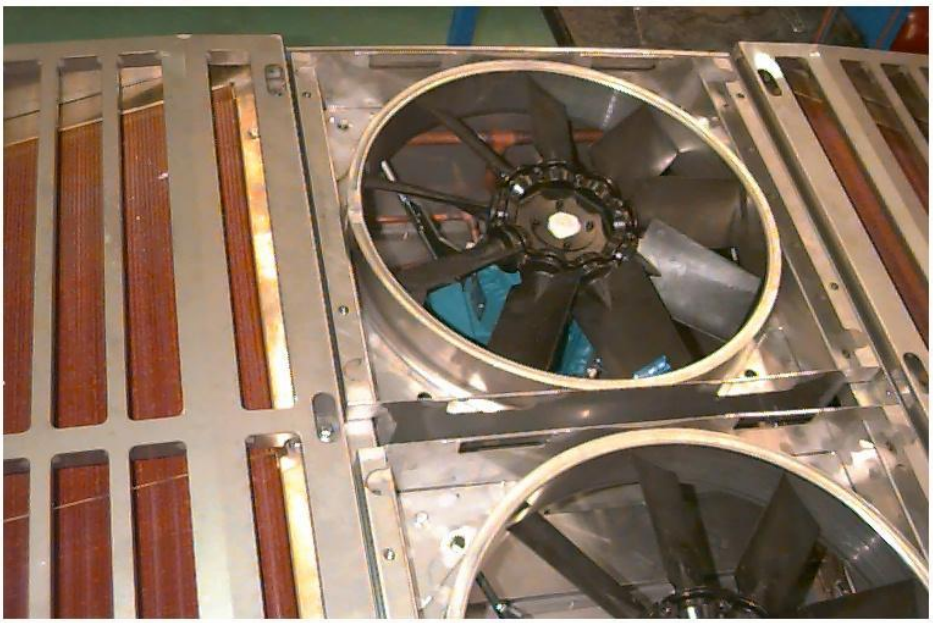
Two multi-blade axial fans circulate the cooling air for the condenser coil. The blades of the condenser fan are of aerofoil geometry, ensuring optimum fan efficiency, and minimum noise.

5.6 Evaporator Coil



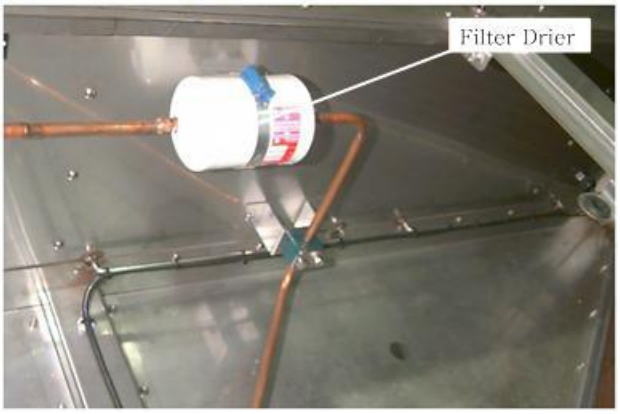
The prime function of the evaporator compartment is to deliver cool air to the main supply air duct that services the vehicle.

5.7 Condenser Coil



Condenser cooling air is drawn in through the condenser coils, mounted on either side of the unit. The condenser air is discharged vertically by the two condenser fans.

5.8 Filter Drier



The filter drier removes dust particles and moisture from the refrigeration system.

5.9 Thermal Expansion (Tx) valve



The TX valve regulates the refrigerant flow into the evaporator coil, expanding the liquid from high pressure to low pressure, and causing a large drop in refrigerant temperature.

6. Doors

A door in a train is a panel or barrier, usually hinged or sliding, that is used to cover an opening in a train.

Exterior doors are doorways that allows entrance to or exit from a train.

There are three types of exterior doors available in RS2:

- Cab or driver's door

- Passenger door

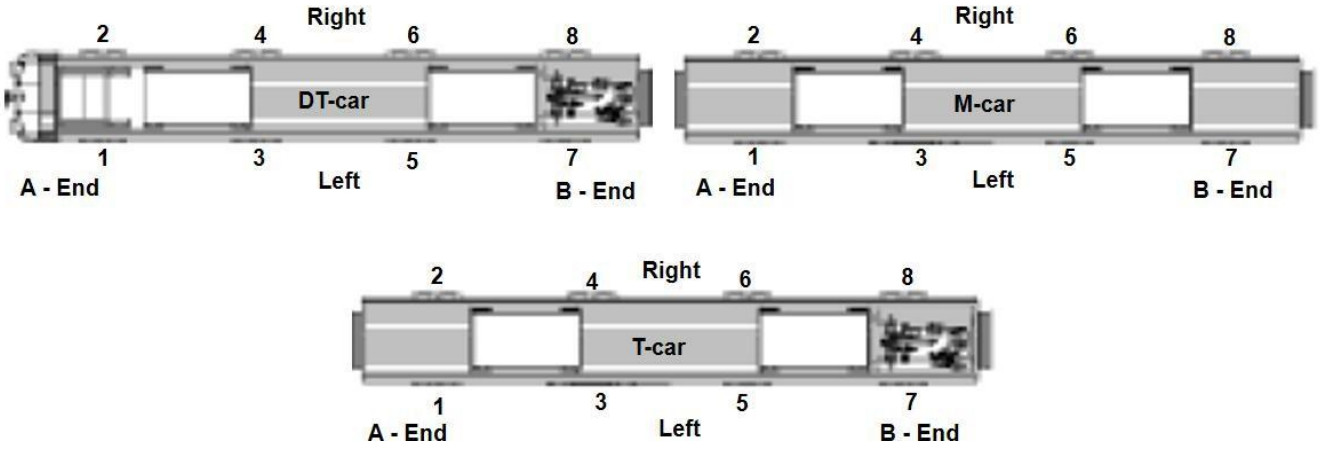
-Detainment door

6.1

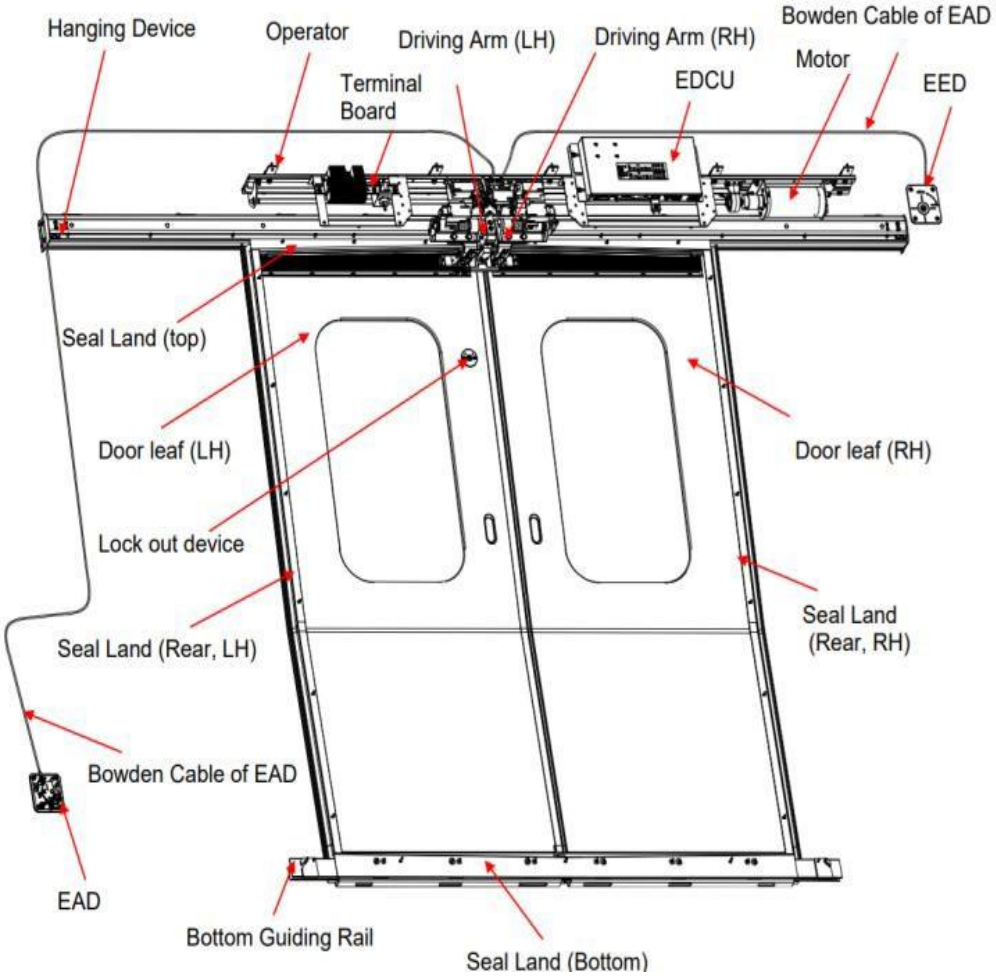


Passenger door

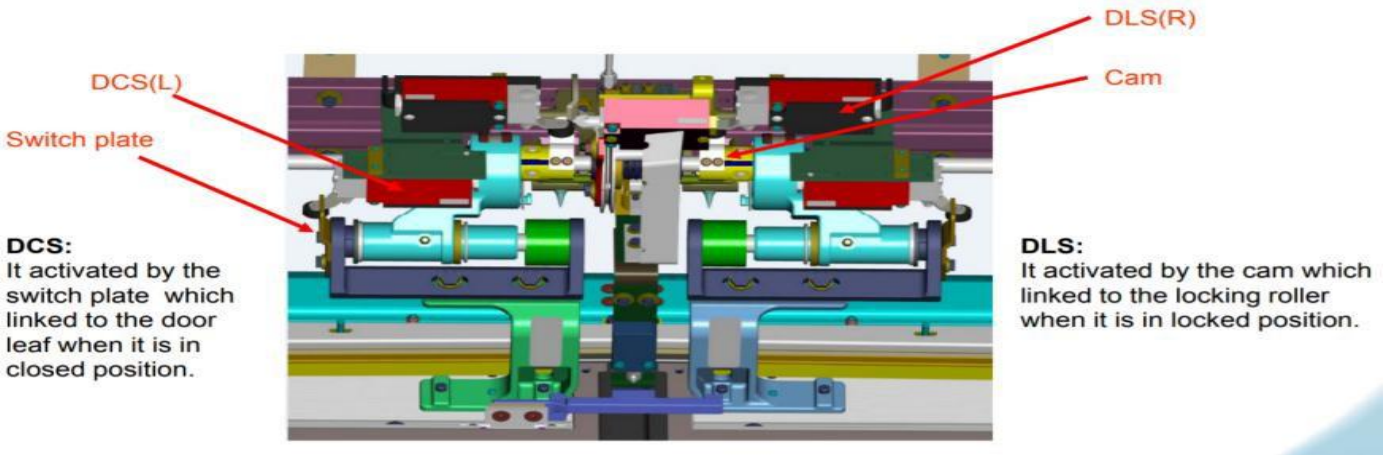
The function of the passenger door is to prevent passenger's from entering or leaving the train, each car has eight passenger doors and consist of two door leaves that sliding externally on the car body wall. Each door is electrically operated. The door leaves are built using stainless-steel sandwich construction with integrated double glazing window panes. Each door leaf is equipped with one external and one internal handle.



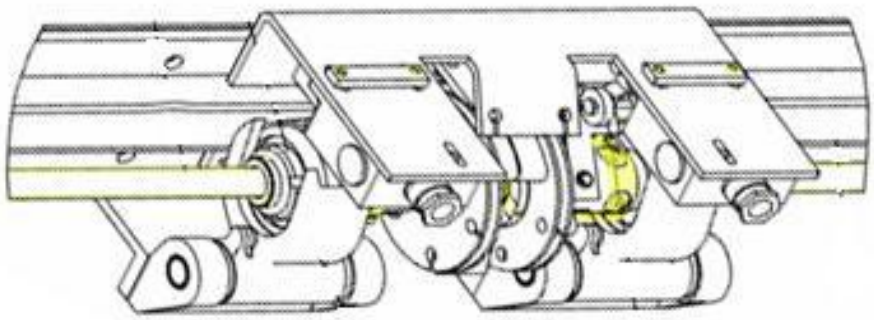
6.2 General door arrangement



6.3 Mechanism Overview



6.4 Drive assembly

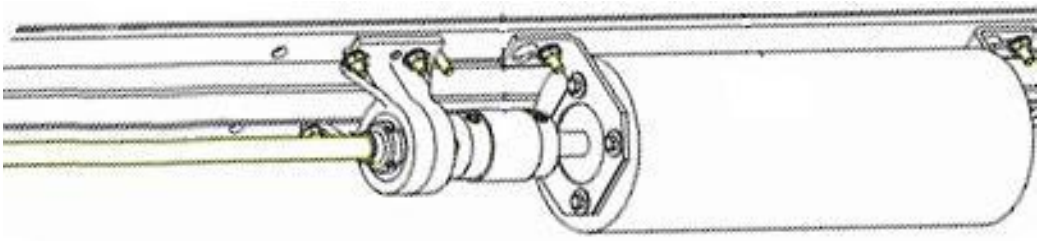


6.5 Motor assembly

Electric motor assembly consist of:

- One electric motor.
- One coupler between motor and driving device.

The driving screw is connected to the motor through an elastic coupling.



6.6 Automatic door locking device

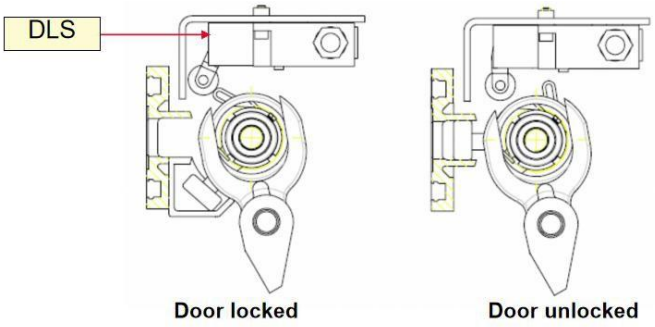
The function of the automatic door locking device is to lock the door after closing, it operates on the principle of preventing the translation of the ball nuts. The unlocking is achieved automatically by rotation of the nut caused by the torque exerted by the motor on the screw during automatic opening.

6.7 Door lock switch

There are two door lock switches (DLS)

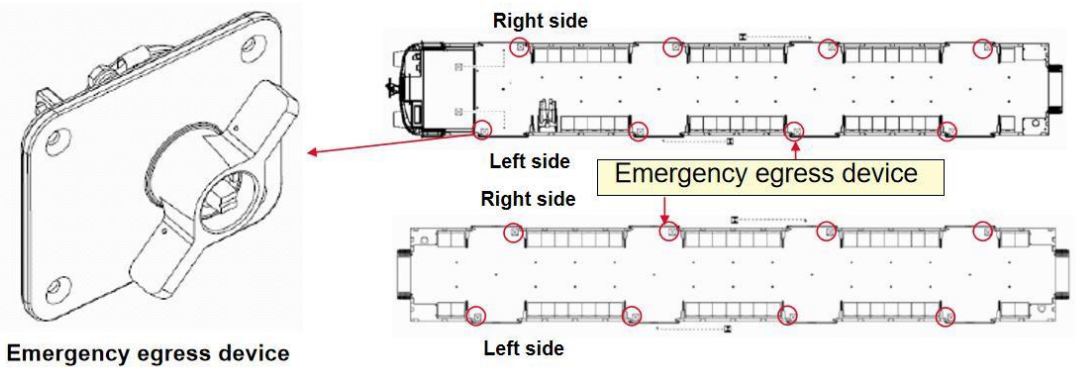
1) To check ball nut rotated into locking groove.

2) To provide door locked signal.



6.8 Emergency egress device

Each door is equipped with an emergency egress device which is mounted on the door machinery. The egress device is accessible through a hatch in the side ceiling cover. It is activated by hand without the use of any tool. It can be reset manually to “normal” position, by turning the handle or the square spindle.



6.9 Electronic Door Control Unit (EDCU)

The electronic door control unit controls the doors fitted on train which is powered by battery voltage. EDCU is having many inputs/outputs. It has a chopper driven PWM controller to control the DC motor. It has maintenance software, system software and an event recorder system.



View publication stats

Conclusion

Delhi metro is a rapid transit system used for mass mobility within Delhi & NCR. A lot of research and engineering is done to make metro eco-friendly and comfortable for modern passenger.

Metro train carrying lakhs of passengers is running smoothly from several years due to various systems in it. Various systems are : Door System, HVAC, Propulsion System, Brakes & Pneumatics, etc, together every system and subsystem is responsible for efficient operation of Delhi Metro.

The bogie is an H shaped structure that supports the the load of the car and the passengers, also various systems are mounted on the bogie like suspension system, braking system etc.

Pneumatic systems plays a major role in the working of the train like applying brakes, blowing horn and raising the pantograph while the train is in dead condition.

Doors are the only way for passengers and driver to get out of the train. The doors are electrically controlled and DCU server as the brain for its efficient working.

All the systems are very well maintained by various depots spread over hectares of land in various parts of Delhi and NCR.

Having constructed a massive network of about 373 Km with 271 stations in record time in Delhi, NCR, the DMRC today stands out as a shining example of how a mammoth technically complex infrastructure project can be completed before time and within budgeted cost by a Government agency.